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Older adults' digital technology experiences: a qualitative study

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Abstract

Background Older persons' use of digital technology is becoming more prominent due to society's heavy reliance on mobile and web applications. A positive attitude has been detected among older persons regarding the use of mobile solutions. Digital technology is increasingly used in the provision of health and social care, which can help older persons continue to be independent, and in turn can contribute to older persons' overall well-being and sense of safety. However, current digital technology has been found to be difficult for older persons. Additional explorations into older persons' experiences using digital technology are therefore relevant.

Method A qualitative design was used with data collected using semi-structured interviews with 17 older persons in the age span 72–101, living in 12 different municipalities. The older persons were admitted to municipality health care. Interviews were conducted in the older persons home, and were audio recorded. The recordings were transcribed and analyzed through a qualitative conventional content analysis. The study method was compliant with the Consolidated Criteria for Reporting Qualitative Research (COREQ) checklist.

Results The analysis resulted in two main categories; *Enrichment, dependance and mixed feelings of using digital technology in daily life* and *Divided preferences and use of digital technology in contact with health and social care*. The first is divided into three sub-categories; *divided usage and feelings about digital technology*, *enriching daily life with digital technology*, and *need for support when using digital technology*. The second is divided into two; *divided experiences regarding digital communication with health and social care* and *a longing for information about healthcare through digital technology*.

Conclusion The diversity showed how older persons cannot be seen as a homogeneous group regarding the use of digital technology. Individual preferences of older persons should be acknowledged by health care professionals and policy makers prior to deciding whether the use of digital technology is preferable, and in which situations. Support and education for older person in using digital technology is essential, as is involving older persons in designing the digital technology that is developed for them to use. Otherwise, older persons may become more dependent on others to manage their everyday life.

Keywords Older persons, Digitalization, Daily life, Healthcare, Social care, Qualitative content analysis

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Background

The growing population of older persons [1] is estimated to double between 2018 and 2050, and the population aged > 80 is expected to triple during the same time frame [1–3]. Healthy aging is defined as the process of developing and maintaining functional ability, which enables well-being in old age, including fulfillment, satisfaction, and happiness in daily living. Functional ability constitutes health-related factors which enable older persons to carry out what they find value in [4]. Older persons experience health or health problems differently; accordingly, policies to promote healthy aging should be framed to improve functionality for all older persons, regardless of whether they are independent or rely on the provision of care [5]. Collaborative efforts between municipalities and healthcare authorities are needed to develop an age-friendly society [6]. The growing population of older persons will affect the composition of society across sectors such as health and social care, housing, and urban planning, [7] as well as the development of digital technology suitable for older persons. However, previous research has reported an exclusion of older persons in the designing of digital technology. An exclusion, which may lead to negative self-perspectives among older persons when struggling to use digital technology [8].

Digital technology, in this study referring to computers, laptops, tablets, smartphones and applications, can change health outcomes by improving medical diagnoses, treatment decisions, person-centered care, and aid in the creation of evidence-based knowledge, if supported by governance to create sustainable solutions [9]. Out of the Swedish population, 85% of persons aged > 75 use the internet, as opposed to over 95% of those aged 55–64 years of age [7]. Older persons' use of digital technology is becoming more prominent due to society's heavy reliance on mobile and web applications [8]. A positive attitude has been detected in previous research among older persons regarding the use of digital technology such as applications [9] and for health reasons, such as contact with healthcare [8, 10, 11]. Research have also described how the increasing use of digital technology the provision of health and social care can help older persons to continue to be independent [12], delivering person-centered care through creating a partnership with the patient [13]. Digital technology can influence person-centered care; improving accessibility, self-management for patients [14], reducing utilization and cost of care, [15] but may also negatively influence person-centered care and the relationship between the patient and the health care professional [16]. The development of digital technology to improve health for older persons can be useful for supporting healthy aging and overall well-being [10, 12, 17] by encouraging older persons to self-manage

health activities and become more independent [12, 17]. However, current digital technology has been experienced to hold difficulties among older persons, such as non-accommodation of physical impairments [18], lack of instructions for how to use applications, high in cost, and lack user-friendly features for older persons [19–22], leading to worry about privacy and confidentiality [10, 19–21], an increased sense of dependence on others when using digital technology [22] and lack of human interaction [10, 20]. Older persons receiving home health and social care often have complex health care needs, where previous research has expressed how the increase of digital technology will increase in home health care [22]. However, previous studies on older persons' experiences using digital technology have mainly focused on other groups of older persons or on health care professionals. The lack of research on older persons receiving health and social care has left a gap of information which need to be deepened. Additional explorations into the experiences of older persons' experiences using digital technology are consequently relevant, and therefore this study aimed to describe the experiences of using digital technology among older persons' receiving health and social care at home.

Method

An inductive qualitative study design drawing from interviews with a semi-structured interview guide was used. The study method was compliant with the Consolidated Criteria for Reporting Qualitative Research (COREQ) checklist [23].

Participants

The heads of the Department of Health and Social Care from 15 municipalities were asked to consent to data collection with patients within their organization, where 12 municipalities consented, and 8 recruited participants. Each municipality appointed a contact person to help identify and inform possible participants orally and in writing in person. Information included the study's aim and method, as well as the researchers' profession and affiliations. If agreeing, the older person's contact information was handed to the researchers. The researcher then contacted the older person and if interest in participating remained, a time and place for data collection was planned. To be included in the project, the older persons had to be ≥ 65 years of age, receive daily services from municipal homecare, and be admitted to municipal home healthcare. The participants of the study were deemed a relevant group to answer the study aim, which strengthened the credibility of the study. Inclusion criteria also include willingness to participate in the study and being able to understand a question and hold a conversation

in Swedish. Seventeen older persons participated in the data collection. Five possible participants declined participation after receiving information by the researchers. Information about possible participants who declined participation after being informed by the contact persons was not relayed to the researchers.

Data collection

This study was a part of a project focusing on evaluating mobile integrated care and data were collected between September 2021 and October 2022. Data were collected through semi-structured interviews performed by the first author. The first author who performed the interviews was a PhD candidate at the time, with experience in performing interviews as well as working with older persons. The second and third author were both associate professors, with extensive experience in qualitative research as well as research including older persons. All were female. There was no relationship between the participants and the researchers. The interviews were guided by an interview guide focusing on several topics, such as experiences with municipal home health care, experiences during the COVID-19 pandemic, and the use of digital technology; the latter of which is the focus of this study. The interview guide had been previously tested in another research project. The questions were open ended, with the goal of capturing the older persons' experiences of digital technology. The participants were asked to participate because of their admittance to municipal home health and social care, and the main focus of the interviews was not the use of digital technology. The participants recruited gave the opportunity to capture a wide range of experience, enhancing credibility of the findings. During the meeting with the older person, the study aim was introduced, and oral and written informed consent were obtained. The researcher first collected demographic data, before starting the interviews, which were audio recorded. The interviews lasted between 19 and 78 min following an interview guide published in this dissertation [24], in the older persons' home per the older person's request. In some cases, the older persons' significant other or other next of kin were present in the home. The section of the interviews pertaining to digital technology contained the following questions: "Tell me of your usage of digital technology," "What are your experiences using digital technology?" "What are your experiences of using digital technology related to your health and social care?" Follow-up questions were asked to further explore participants' experiences. After the interviews, field notes were written by the interviewer, which were not used in the analysis but worked as a reflective tool to promote discussion in the research group. The

transcripts were not returned to the participants, but the contents of the interview were discussed with the older persons when the researcher came back for a second data collection in the bigger project.

Data analysis

Data analysis was conducted through the steps of conventional content analysis, which was chosen since it can be used to describe a phenomenon inductively. The analysis method is suitable for the aim of the study of describing the experiences of using digital technology among older persons' receiving health and social care at home. The choice of the analysis method was further motivated by the possibility of gaining direct information from an under-researched group in the field of digital technology. The study participants' experiences could therefore be presented in the findings without imposing theoretical perspectives on the analysis [25]. The choice of analysis was appropriate in relation to the aim. The analysis followed the steps described by Hsieh and Shannon. First, all the researchers read the interview text regarding experiences using digital technology repeatedly, to achieve immersion and gain a sense of the whole. Secondly, the data were read word-by-word, to isolate meaning units by highlighting the words which appeared to capture key thoughts or concepts related to the study's aim. This was conducted by the first author. Thirdly, the meaning units were condensed into shorter units which held the same meaning. The condensations were then given a code. The process was followed by a discussion in the research group about the condensations and codes. Afterwards, notes about the first part in the analysis (thoughts and impression) were made by the research group, indicating two main groups in the findings: digitalization in daily life and digitalization in relation to health and social care. These notes were discussed in the research group to reach consensus. After this, the codes were sorted into sub-categories based on how the codes were related and linked to each other. These codes were then organized and grouped into meaningful clusters within the sub-categories. These were once again discussed in the research group to reach consensus. Smaller sub-categories were then combined and placed in a hierarchy which created the main categories (Table 1). The names of the sub-categories and categories were reached by discussion in the research group to reach consensus. The data was analyzed manually, and no software was used to conduct the analysis. The study finding's trustworthiness was demonstrated through credibility, dependability, and transferability [26], which is discussed in the 'strengths and limitations' section.

Table 1 Example of the analysis process

Meaning unit	Condensation	Code	Sub-category	Category
I have this smartphone, and all my children, grandchildren and great grandchildren can call and show me what they're doing	Using digital technology to keep in contact with loved ones	Digital technology as a line to loved ones	Enriching daily life with digital technology	The enrichment, dependance and mixed feelings of using digital technology in daily life

Ethical considerations

The project was conducted according to the Declarations of Helsinki [27] and approved by the Ethical Review Authority (Dnr 2020–07149). The ethical approval includes approval for recruitment and research plan containing a section on ethical considerations. Participants received oral and written information about the project and the voluntary nature of their participation, as well as the ethical considerations of the project. Written informed consent was obtained from all participants.

Findings

Seventeen older persons participated in the data collection (Table 2).

The findings are represented in two main categories, the first of which has three sub-categories (Fig. 1) and focuses on digital technology use in the daily life of the older person. The second main category has two sub-categories and describes the experiences of older persons using digital technology in contact with health and social care. The term ‘healthcare personnel’ will hereafter be used when referring to all groups of personnel involved in health and social care. When referring to personnel

with a healthcare certificate, the term ‘healthcare professionals’ will be used.

Enrichment, dependance and mixed feelings of using digital technology in daily life

The experiences of the older persons’ in this study use of digital technology in daily life was divided. Some participants used digital technology on a daily basis. There were also those who described not using digital technology at all. There were older persons who felt frustrated or frightened by digital technology, while other felt comfortable and proud about using it. Digital technology was described as enriching the participants lives, as they could keep in contact with their friends and families, and keep up with hobbies. Furthermore, the older persons described sometimes needing support with their digital technology, and younger relatives were the primary source of support.

Divided usage and feelings about digital technology

There were divided experiences using digital technology among the older persons participating in the study. When asked about digital technology, some of the older persons were unsure what constituted digital technology. Among those who used digital technology, the older persons described using smartphones or tablets. There were also experiences of previously having used a stationary computer during their work life, but none of the older persons participating in this study used them currently. Other participants described never using digital technology, referring to having a corded phone or a regular cellphone (not a smartphone). Furthermore, there were experiences of having physical impairments which hindered the older person from using digital technology, such as crooked fingers, or hearing- or vision loss.

Table 2 Characteristics of participants

	Patients (n = 17)
Age range (mean)	72–101 (85)
Female (n)	9
Male	8
Retired	17
Children	17
Grandchildren	17
Co habituating partner	5
Living alone	12

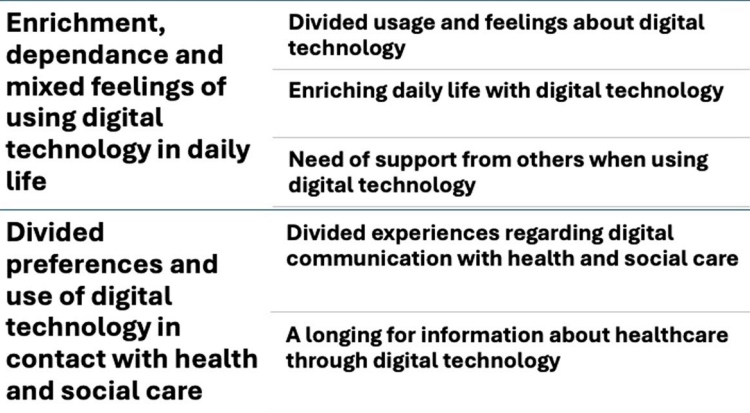


Fig. 1 The findings structure of categories and sub-categories

The older persons participating in this study had divided feelings regarding the use of digital technology. Some did not feel comfortable using digital technology at all, and did not see it as an option in daily life. One older person said: *"No, those don't come into this household (N2)"*. Furthermore, there were descriptions of digital technology as frightening, with the older persons referring to how much money digital technology cost. Choosing not to use digital technology was at times described by the older persons as being 'old fashioned' and some explained that they had never liked digital technology. One said: *"It has never really been my thing, the technical stuff. (N16)"* In contrast, there were those who felt very secure with their cellphones, and did not see any reason to use anything new. One said: *"If my kids got to decide I'd have one of those tablets, but I refuse! (N5)"* There were also experiences among the participants having grown into using digital technology. Those who had not had any experience with computers when they were younger now used tablets on a daily basis. For some who had used computers during their work life, found it easy to switch to using tablets. One said: *"It works really well. I'm very glad for it.(N3)"*

Enriching daily life with digital technology

Digital technology was described as having enriched the older persons' lives. The older persons participating in this study used digital technology such as video meetings to keep in contact with their family and friends. However, there were also those who had no experience using video meetings as a way to interact with others. The older persons described having started using digital technology during the COVID-19 pandemic to decrease the risk of becoming infected with the virus, but afterwards continued to use digital technology in a way which enriched the older persons' life. One said: *"I have this smartphone, and all my children, grandchildren and great grandchildren can call and show me what they're doing. (N17)"* There were also experiences of having had church or group meetings through video meetings. Learning to participate in video meetings took time, but the older persons participating in this study could do it themselves. There were descriptions of having video meetings often, while others had only tried it a couple of times. The older persons also sent texts or images to family and friends, and had family and friends send them text messages and images in return. It was further described how the participants had previously been able to send images in the past, but was no longer able to. One said: *"I used to take photos of my flowers and send them to my daughter, but I can't remember how to anymore. (N2)"* Digital technology was also used for other daily tasks. The older persons used their digital technology to read the news, to watch

movies and television, and to make purchases. One said: *"and I bought vacuum bags too! They're always so difficult to find in the store, the right ones, but there's a whole webpage about it. (N3)"* Digital technology was also used to remind the older persons to take their medications. Some of the participants paid their bills digitally, while others had no experience doing so.

Need of support from others when using digital technology

The older persons participating in this study described being supported by their adult children, their children's spouses, and their grandchildren in the use of their digital technology. These relatives helped the older persons set up video meeting programs, read their documentation online, order medications, and pay bills. The older persons experienced that there was never an issue getting help from their family and friends. One said: *"There's never any trouble, they're so sweet to help me with it all. (N16)"* Other older persons participating in this study had experiences with a digital coach, who was an assistant nurse employed by the municipality to support older persons in using digital technology. The digital coach had helped the older persons learn how to make a video call, which applications on the tablet which might be useful, and in finding music videos of their favorite artists. One said: *"and then she helped me find Bruce Springsteen, who is a big idol of mine. (N11)"*

Divided preferences and use of digital technology in contact with health and social care

There were differing experiences described by the participants regarding whether they had used digital technology in contact with health and social care. There were older persons participating in the study who experienced having used digital technology in contact with health and social care, such as participating in video meetings or reading the documentation made by healthcare online. Others had no experience using digital technology in contact with health and social care. Furthermore, the older persons described how digital communication between healthcare personnel both improved their health and social care, but also made them feel invisible.

Divided experiences regarding digital communication with health and social care

There were divided experiences regarding digital communication with different organizations among the older persons. Some had experiences of video meetings with healthcare professionals from the hospital, but the municipal healthcare to which the participants had been admitted had not requested any video meetings. Video meetings with the hospital had taken in one of two forms.

The first was health and social care planning meetings when the older person was to be discharged from the hospital, where the municipality's registered nurse and sometimes their next of kin participated through video, while the older person and hospital professionals were in the same room at the hospital. The second type of video meetings were continuous checkups with the hospital's healthcare professionals, when the participant was at home. Video meetings with healthcare professionals from the hospital were generally described by the older persons as having worked well. The older persons participating in this study described not having had contact with municipal healthcare through video meetings, but would prefer video meetings over the phone communication they currently experienced with the municipality's registered nurse.

Several different professions participated in the health and social care plan meetings. This led to it being impossible to distinguish the professions in relation to the persons in the meeting room. One said: *"It really was impossible to know who was who around the table, or who was even there. (N1)"* The older persons described mostly listening during the meeting, with little possibility to participate in what health and social care they would need when discharged. Others described that they had their questions answered during the meeting. The older persons participating in this study felt the hospital was professional in using video meetings, and felt that some they were experienced in using a digital medium for discharge meetings with the municipality. Other participants relayed experiences of the technology malfunctioning. When the municipality's registered nurse who was responsible for the older persons' care, participated in the video meeting, the older persons participating in this study felt seen and experienced security in having someone present who knew them. However, others described how the municipal healthcare professionals looked disinterested during the video meeting. One older person said: *"She seemed so uninterested, like she didn't care at all about me. (N11)"* The older persons also described how they felt too ill to have the energy to participate in healthcare planning through digital technology in the way they wished to. Hospital professionals were described as unfamiliar with who the older person was, opposed to the municipal healthcare professionals, which made the older person feel more exposed during the video meeting. Furthermore, the participants experienced a lack of time during the video meeting, which made the meeting feel rushed and hindered communication.

Video meetings when the participants had to follow-up treatment from the hospital were conducted while the older person was in their home, using a tablet or smart phone. The video meetings were appreciated, since it

saved time for the older person, especially for those who had a long way to travel. One said: *"It's shit that you have to travel so far to the hospital for a fifteen-minute check up with someone who just disagrees with you, and then it's over. It's much better to do that over the phone or video. (N4)"* They also described it as easier to involve next of kin in these meetings, since they could participate regardless of where they were living. Older persons participating in this study felt that healthcare could increase the use of video meetings, and that healthcare professionals should be well acquainted with using digital technology to a larger extent by now to help patients save time. The older persons stated that some healthcare professionals were not comfortable using video meetings for follow up treatment, because it was still new to them. Having video meetings was experienced by the older persons participating in this study as not being as supporting as in-person meetings, and in the case of bad news, the older persons wished for an in-person meeting.

A longing for information about healthcare through digital technology

The older persons experienced not knowing where to get digital information about what care they were entitled to in terms of health and social care. They felt they had not received any information about where to find such information. Remembering information was a struggle, and a place to read about one's own care was something the older persons who participated in this study wished for. One said: *"I would like a place where I can read about my own care in real time, and go back to read as well afterwards. (N6)"* The majority of the older persons participating in this study had not read the healthcare documentation available online. There were older persons participating in this study who were unaware that their documentation was available online and wondered whether they could read all their documentation there. Several had considered reading the documentation. Other participants had next of kin who had read their documentation and told them about it. Furthermore, there were experiences of having read printed documentation, which had been handed to the older person by the municipality or by the hospital, but was not available online. The older persons who had read the documentation online found it to be a good service, but were frustrated about how they could only read the hospital and primary healthcare documentation, and not the municipality's documentation. Being able to read the hospital and primary healthcare documentation was considered as a way of clarifying information. It was a good way to go back and try to understand the information which had been difficult to grasp at a healthcare meeting. One said: *"I couldn't understand what the doctor meant, so I logged*

in to read it instead. (N4)" The older persons also had experiences reading more information than they wanted to know online, information they had not yet been informed about, which made them uncomfortable. Furthermore, participants described wanting a place where they themselves could document their health, as well as read the documentation which the municipality personnel had written about them. One said: *"It would be good if it could be like a calendar, where I could write how I'm feeling, but read what they're writing too. (N11)"* Being able to access information about their healthcare through digital technology could make them feel more included in their health and social care.

Discussion

The older persons in this study described divided experiences of using and feelings towards digital technology, where some were positive about digital technology and used them daily, some were hesitant, and others did not use them at all. The differences in attitudes towards digital technology could be viewed through the lens of the Technology acceptance model (TAM) [28], which describes how the attitudes of using technology is based on the perceived usefulness and the perceived ease of use. The inductive findings of the present study will be discussed against the framework by Davis in 1989, as well as previous research. There were obstacles found in relation to using digital technology from the participants. The older persons in this study experienced digital technology as sometimes frightening and difficult to understand, which made them feel 'old fashioned,' and could be related to the perceived ease of use in TAM. Since the design of digital technology often omits including older persons, their view of themselves as being 'old fashioned' and not technical might be fulfilled. Previous research has found this as well, describing how obstacles such as fear hindered older person from using digital technology, which may be because digital interfaces for older persons is not well applied within designs [21]. Furthermore, the older persons in this study, as well as previous research [21], described having functional impairments which hindered them from using digital technology. Being unable to use digital technology leads to digital exclusion [29]. This too could be viewed in the light of the perceived ease of use [28]. In relation to healthcare, the older persons in the current study described frustration with lacking access to the documentation from the municipality digitally. The frustration at the lack of joint documentation between health care authorities was shared by municipal home healthcare professionals in previous research, who also struggled with having to navigate different documentation systems [30, 31]. Previous research has shown how access to information digitally is beneficial for older

persons' health [32–34]. The older persons in this study expressed a wish to document themselves in the digital documentation systems, which could be an development of patient participation and increase the possibility of creating a partnership with the patient and improve person-centered care. It could also add to the perceived use of the digital technology. In the findings there were older persons who had not read their documentation online, or did not know it existed. The lack of using digital technology offered could suggest that the older persons did not perceive it to be useful, in accordance with TAM [28]. Not knowing it existed could instead be an example of a digital divide between the older person participating in the current study and other groups [35].

Positive aspects regarding the use of digital technology were expressed by the older persons participating in this study. Having digital technology was described by older persons in this study as enriching their daily lives, as keeping in contact with family and friends and participating in activities became easier. This would be in accordance with how TAM expresses attitudes towards digital technology being influenced by the usefulness, and should be acknowledged when digital technology is suggested for older persons to use. Previous research has shown how, during the pandemic, social distancing was practiced to protect older persons from becoming infected [36], which led to social isolation and loneliness [37], as well as declined health for older persons admitted to municipal home healthcare [38]. In contrast, the older persons in the current study described how digital technology had aided them during the pandemic, enabling social connections. Developing digital technology like applications which promote social interactions with friends, family members, senior centers, and churches have been shown to be of great significance when using digital technology to improve social connectedness [39] for older persons who may experience involuntary loneliness. However, digital technology cannot fully replace face-to-face connection, but can be a supplement [29], and has, in other research, been argued to alleviate loneliness [40]. There were older persons in the present study who described never having used digital technology when communicating with family and friends, relying instead on their phones. For those who struggle with digital technology, hybrid solutions where one can participate both in person and digitally have been found to help facilitate participation in social activities [39]. The older persons participating in this study described having their adult children support them in using digital technology, especially when there was something the participants struggled with. Support from others is needed for older persons to aid in the perceived ease of use. Previous research has shown a gap between the potential benefit

of digital technologies and the barriers older persons face in the adaptation of digital technology [41], where infrastructure development and awareness of digital literacy, especially for older persons, needs to be addressed [42]. While the older persons in the current study did not describe any issues receiving help from their relatives, education for older persons can provide meaningful improvements in the development of digital technology use, improving aging literacy and limit the digital divide. Co-production of the design phase of digital technology may improve its usage among older persons [41], which could improve the attitudes of older persons if managing to create the a perceived usefulness and ease of use. Furthermore, municipality health and social care workers—such as the digital coaches mentioned in the findings, may be effective informational messengers for older persons [39]. Digital communication holds the promise of building digital skills, improved health outcomes [43], and increased access [44]. Moreover, older persons should be educated on digital technology which may benefit older person [39].

The older persons in the present study described how digital technology had improved their healthcare, since it made communication between healthcare providers easier. This finding stands in contrast to some previous research, claiming digital technology can negatively influence person-centered care and the relationship between health care professionals and the patient [24]. The communication aspect in regard to digital technology is of significance, since healthcare is information-intensive, and timely, accurate, and relevant information can be the difference between life and death [39]. The older persons in the present study described how they experienced different digital technology such as smart phones and cameras to increase the quality and swiftness of their care. Previous research describes how a paradigm shift has been instigated through digital transformation of healthcare [45]. However, the older persons participating in the study described differences in using digital technology the among healthcare professionals, as some healthcare professionals seemed comfortable using digital technology, while others had had the opposite experience. The attitudes of healthcare professionals have previously been shown to influence the acceptance and efficiency of the use of digital technology in practice, and should therefore involve older persons in an early stage of digital technology design [46]. The older persons in the present study found video meetings to work well, but none had experienced municipal healthcare personnel initiating video meetings. Healthcare professionals working in municipal home healthcare confirmed this in their descriptions of using phone calls and not video meetings to keep in touch with patients [38]. In the findings, the older

persons described how access to information through digital technology was positive. Information could at times be difficult to remember or understand, and having access online could help them recall information. The perceived usefulness of technology could therefore be seen as a driving factor in wanting to use digital technology, along with the attitudes of the older persons. These are important findings to take into account when tailoring digital technology to fit the older persons' needs.

Strengths and limitations

The aim of the study was to describe older persons' experiences using digital technology. The study finding's trustworthiness was demonstrated through credibility, dependability, and transferability [26]. The participants of the study were deemed a relevant group to answer the study aim, since they were persons >65 years of age who were receiving home health and social care, and thereby strengthened the credibility of the study. The advantage of conventional content analysis is gaining direct information from the study participants [25], which allowed for describing the older persons' experiences. The participants were asked to participate mainly because of their admittance to municipal home health and social care, and the main focus of the interviews was not the use of digital technology. The participants were informed prior to the study that questions about digital technology would be included; this could indicate that the participants did not have a particular bias regarding digital technology, which could have contributed to the diverse findings as well as capturing a group which would otherwise not have participated in a study on digital technology. Some of the participants were found to lack experience of using digital technology, which could be viewed as a limitation of the findings as well as a limitation in the group chosen to participate. The lack of experience among the participants were however a finding which was important in itself, showing the diversity in the group. In future research, exploring the phenomenon with participants who have varied demographics and contexts would further deepen the knowledge of the phenomenon.

Data were collected through semi-structured interviews where the participants were asked the same questions. The follow up questions varied questions depending on the participants responses. Interviews can aid in gaining a deeper understanding of a phenomenon, which was the intention of the study [47]. The participants themselves decided where and who were present during the interviews. Having next of kin present during the interviews may have influenced the older persons responses. The participants themselves chose the setting of the interview they were considered comfortable and hopefully created no bias. The content

analysis method aims to attain a condensed and broad description of the phenomenon in which the outcome of the analysis results in concepts, categories or themes [48]. It is important to perform peer debriefing and prolonged engagement to strengthen the study's credibility [49]. While peer-debriefings were conducted within the research environment, prolonged engagement were not and could be seen as a limitation. However, none of the participants were perceived to have felt uncomfortable to express their experiences of digital technology. Furthermore, a limitation of the study could be seen as the participants not having the opportunity to provide feedback on the results. Their responses from the interview were however discussed during another visit. Dependability was strengthened through the continuous discussion about the study findings in the research group through the analysis. Transferability relies on the readers ability to determine whether the findings are transferable, which relies on the thorough description of the participants, the research context and the analysis process [47]. Older persons using digital technology is a worldwide phenomenon through the digitalization of society, and the findings could therefore be transferable to several settings. Transferability of the findings is up to the reader and is made possible by the thorough description of the participants, data collection, and analysis.

Conclusions

Overall, there were differences in older persons' experiences using digital technology in daily life and in communication with health and social care, demonstrating the diversity in experiences of the group. The findings were in accordance with TAM, where the attitudes of older persons participating in the study were influenced by the technology's perceived ease of use and perceived usefulness. In consistency with previous research, there were older persons who expressed an enrichment in their life, but also those who found digital technology frightening, and those who were physically unable to use it and or refused to use it because they did not want to. Older persons can therefore not be seen as a homogeneous group in their experiences and preferences regarding using digital technology according to the present study as well as previous research. The divided descriptions show a range of experiences among older persons, where health care professionals cannot assume all older persons have the same preference when using digital technology in terms of communication with healthcare personnel. It is therefore of importance to tailor the procurement and use of digital technology to the diverse needs of the older persons. The individual preference of older persons is therefore recommended to be acknowledged by health care professionals and

organizations, prior to deciding whether the use of digital technology is preferable, and in which situations. Health and social care organizations should integrate older persons in the design and implementation of digital technology, to ensure they are useful and usable.

The older persons in this study expressed that support in using digital technology is essential, and that the support often came from younger relatives or from a hired digital coach. It is important to provide education to older persons and to involve them in designing the digital technology they are planned to use. Otherwise, older persons may become more dependent on others to manage their everyday life. A finding which had not been found in previous research, was how older persons experience lack of participation in the documentation of their health and social care. For older persons to be able to do so, developing a partnership with the patient and improve the strive towards person-centered care should be improved.

Abbreviation

TAM Technology acceptance model

Glossary

Digital technology	computers, laptops, tablets, smartphones, applications
Smart phone	a cellular telephone with an integrated computer and other features not originally associated with telephones, such as an operating system, web browsing, and the ability to run software applications
Tablet	a wireless, portable personal computer with a touch-screen interface. The tablet form factor is typically smaller than a notebook computer, but larger than a smartphone
Cellphone	a wireless handheld device that allows users to make and receive calls. In this paper a cellphone is a phone with buttons and no touch screen
Video meetings	live, visual connections between two or more remote parties over the internet that simulates a face-to-face meeting. Video conferencing is important because it allows people who would not normally be able to have a face-to-face connection to experience one remotely

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

L.E, J.H and C.G designed the study, contributed to the data analysis, the intellectual content of the manuscript and reviewed the manuscript. L.E collected the data and drafted the manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Swedish Ethical Review Authority, Sweden (Dnr 2020–07149) and was in adherence with the Declaration of Helsinki. All participation was voluntary, and informed consent were obtained both orally and in writing.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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